

Asset and Liability Management Project
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Prepared by :

DIENDERE Wend Kouni Oliseh

KAMGA BOPDA Davy Romaric

ZITAN Layla

Program :

MSc. Actuarial Science, LSBA

Professor :

Jérôme BARBARIN

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Introduction

Ce travail examine la gestion actif-passif d'une compagnie d'assurance-vie sous deux angles complémentaires. La question de fond est simple : lorsque les taux bougent, qu'arrive-t-il aux fonds propres ? La première partie répond à cette question de manière déterministe — bootstrapping de la courbe zéro-coupon, valorisation des provisions techniques, analyse du gap de duration et couverture par swap à départ différé. Les outils sont classiques, mais leur application au bilan d'un assureur-vie révèle d'emblée une asymétrie structurelle : les passifs sont notablement plus longs que les actifs. La seconde partie repose sur le modèle Hull-White G1++ et des simulations Monte-Carlo sous la mesure risque-neutre. Elle introduit la participation aux bénéfices comme une option imbriquée dans les provisions, puis reconstitue la distribution du P&L sous la mesure historique pour calculer le SCR selon Solvency II. L'objectif n'est plus seulement de mesurer le risque, mais d'en comprendre les sources et les leviers de réduction.

Chapitre 1

Deterministic Model

Question 1 : Yield Curve Construction via Bootstrapping

We construct the risk-free zero-coupon curve using a standard bootstrapping algorithm from market instruments :

- **Short-end 0–1 year** : Deposit rates and *Ester* overnight rate provide discount factors via simple compounding :

$$P(0, T) = \frac{1}{1 + R_{\text{dep}} \times T}, \quad T \leq 1 \text{ year.}$$

- **FRA segment** : The 6×12 FRA rate R_{FRA} implies the forward discount factor :

$$P(0, 1) = P(0, 0.5) \times \frac{1}{1 + R_{\text{FRA}} \times 0.5}.$$

- **Swap curve** : For each maturity T_i , we solve iteratively for $P(0, T_i)$ such that the fixed-rate swap equals par :

$$\text{Swap Rate}_i = \frac{1 - P(0, T_i)}{\sum_{j=1}^{i-1} \tau_j P(0, T_j)},$$

where τ_j is the year fraction. This yields the zero-coupon price $P(0, T_i)$ and the continuously compounded spot rate : $r(0, T_i) = -\frac{\ln P(0, T_i)}{T_i}$

Market Inputs and Results

The constructed curve exhibits an upward slope from 1.36% (1Y) to 2.43% (20Y), consistent with a normal yield curve environment. The bootstrapping ensures exact calibration to market swap rates, providing an arbitrage-free discount curve for pricing the insurance liabilities.

Maturity (years)	Swap Rate	Discount Factor $P(0, T)$	Zero Rate (Cont.) (%)
1	–	0.986	1.36
2	1.304%	0.974	1.30
5	1.371%	0.934	1.36
10	1.761%	0.838	1.77
15	2.112%	0.724	2.15
20	2.356%	0.615	2.43

TABLE 1.1 – Bootstrapped zero-coupon curve from market instruments.

Question 2 : Mathematical Statutory Reserve

The mathematical reserve (MR), also known as the technical provision, represents the present value of the insurer's future obligations to policyholders. It is computed as :

$$\text{MR} = \mathbb{E}^{\mathbb{Q}} \left[\sum_{i=1}^7 \frac{N_i \cdot Cl_i(T_i) \cdot {}_{T_i}p_{x_i}}{B_{T_i}} \right], \quad (1.1)$$

where :

- N_i is the notional amount for model point i ,
- $Cl_i(T_i)$ is the guaranteed capital at maturity T_i (including profit-sharing bonuses),
- ${}_{T_i}p_{x_i}$ is the survival probability under the Makeham mortality law,
- B_{T_i} is the discount factor from the money-market account.

Numerical Result

Component	Value (€)
Total Mathematical Reserve (MR)	1 905 003 947.92

TABLE 1.2 – Total mathematical statutory reserve across all model points.

Question 3 : Economic Value and Impact of Surrender

Economic Value Calculation

The economic value (EV) of liabilities is computed as the risk-neutral expectation of discounted cash flows, incorporating the profit-sharing mechanism and survival probabilities. We evaluate the impact of the surrender option, which allows policyholders to terminate contracts early, with surrender value $SV(t) = Cl(t)$ (no penalty) and surrender allowed from $t = 1$ to $T - 1$.

Comparison with Mathematical Reserve

Valuation Metric	Value (€)	Difference vs MR
Mathematical Reserve (MR)	1 905 003 947.92	–
Economic Value no surrender	1 538 320 632.70	–19.2%
Economic Value with surrender	1 545 536 472.71	–18.9%

TABLE 1.3 – Comparison of valuation approaches.

The mathematical reserve exceeds the economic value by **366.7 M€**. This gap arises because the mathematical reserve uses **regulatory discount rates**, which are often more conservative, whereas the economic value employs **market-consistent risk-free rates** derived from the Hull-White model. Additionally, the mathematical reserve may include **prudent margins** for adverse deviation, while the economic value reflects best-estimate expectations under the risk-neutral measure.

The surrender option increases the economic value by :

$$\Delta EV = 1,545.5 - 1,538.3 = \mathbf{7.22 \text{ M€}} \quad (+0.47\%).$$

This slight increase occurs because $SV(t) = Cl(t)$ (no penalty) and, when the guaranteed rate g is lower than the risk-free rate r , the surrender value can exceed the discounted value of waiting : $Cl(t) \gtrsim Cl(T)/(1+r)^{T-t}$. This makes early surrender marginally more costly for the insurer, particularly near maturity.

Model Point Heterogeneity : The surrender impact varies across model points. The aggregate effect remains limited +0.47%, indicating that the surrender option is **nearly neutral** under the current assumptions.

Although the impact is small, the surrender option introduces behavioral risk that requires monitoring under Solvency II (SCR for lapse risk). A modest surrender penalty, 1–2%, would render the option financially neutral for the insurer while preserving policyholder flexibility.

Question 4 : Market Valuation of the Coupon Bond

The asset portfolio consists of three fixed-rate coupon bonds with maturities of 3, 5, and 8 years. Each instrument is valued by discounting its future cash flows using the risk-free Nelson-Siegel curve augmented with a bond-specific credit spread. The dirty price is computed as the sum of discounted cash flows per unit of nominal, and the total market value aggregates the individual positions. This market-consistent valuation provides the accurate asset base required for duration gap analysis and solvency assessment.

Bond	Mat. y	Coupon %	Spread %	Nominal M€	Dirty Price	Market Value M€
1	3	2.40	1.50	550	94.26	518.43
2	5	3.30	2.70	670	87.84	588.52
3	8	4.80	2.20	720	92.70	667.45
Total Portfolio				1 940		1 774.39

TABLE 1.4 – Market valuation of the coupon bond portfolio.

All three bonds trade at a discount (dirty prices < 100), reflecting that their coupon rates are below the corresponding risk-free plus credit spread yields. The total portfolio market value amounts to **1.774 Bn€**, representing a discount of **8.5%** relative to the aggregate nominal of 1.940 Bn€. This mark-to-market adjustment is critical, as it reduces the effective asset base available to cover technical provisions and directly influences the duration gap and equity sensitivity calculations.

Question 5 : Duration and Convexity of Technical Provisions

The duration and convexity of the technical provisions are computed as the first and second discounted time moments of the liability cash flows :

$$D_{TP} = \frac{\sum_i T_i EV_i}{\sum_i EV_i}, \quad Q_{TP} = \frac{\sum_i T_i^2 EV_i}{\sum_i EV_i}.$$

The corresponding dollar measures are :

$$\$D_{TP} = D_{TP} EV_{TP}, \quad \$Q_{TP} = Q_{TP} EV_{TP}.$$

Quantity	Without Surrender	With Surrender
Economic Value of TP	1 538 320 633	1 545 536 473
Dollar Duration	11 238 359 845	11 373 895 233
Duration	7.3056	7.3592
Dollar Convexity	108 324 443 355	110 084 896 164
Convexity	70.4173	71.2276
Dispersion	17.0455	17.0700

TABLE 1.5 – Duration and convexity of the technical provisions.

The duration of the technical provisions is around 7.31 years without surrender and 7.36 years with surrender. The surrender option slightly increases the economic value, duration and convexity of the liabilities. Under the no-penalty assumption, surrender generates additional intermediate payments and therefore slightly increases the insurer's liability value and interest-rate sensitivity.

Question 6 : Duration and Convexity of the Bond Portfolio

For each bond, cash flows are discounted using the Nelson–Siegel risk-free rate plus the credit spread :

$$P_j = \sum_t CF_{j,t} e^{-r_j(0,t)t}, \quad r_j(0,t) = r_{NS}(0,t) + s_j.$$

The duration and convexity are :

$$D_j = \frac{\sum_t t PV(CF_{j,t})}{P_j}, \quad Q_j = \frac{\sum_t t^2 PV(CF_{j,t})}{P_j}.$$

Portfolio duration and convexity are market-value weighted averages :

$$D_A = \sum_{j=1}^3 w_j D_j, \quad Q_A = \sum_{j=1}^3 w_j Q_j, \quad w_j = \frac{MV_j}{\sum_k MV_k}.$$

Bond	Market Value (€)	Weight	Duration	Convexity
Bond 1	518 430 067	29.217%	2.9274	8.6861
Bond 2	588 515 755	33.167%	4.6621	22.6548
Bond 3	667 445 916	37.616%	6.7553	50.5236
Total Portfolio	1 774 391 738	100.000%	4.9426	29.0565

TABLE 1.6 – Duration and convexity of the bond portfolio.

Bond 3 has the highest duration and convexity because it has the longest maturity. The asset portfolio has a total market value of 1 774 391 738 €, a duration of 4.9426 years and a convexity of 29.0565.

Question 7 : Duration Gap and Convexity Gap

The dollar duration and convexity gaps are computed as :

$$\$D_{Gap} = D_A V_A - D_L V_L, \quad \$Q_{Gap} = Q_A V_A - Q_L V_L.$$

The relative gaps are expressed with respect to equity :

$$D_{Gap} = \frac{\$D_{Gap}}{S_0}, \quad Q_{Gap} = \frac{\$Q_{Gap}}{S_0}, \quad S_0 = V_A - V_L.$$

Quantity	Without Surrender	With Surrender
Asset Value V_A	1 774 391 738	1 774 391 738
Liability Value V_L	1 538 320 633	1 545 536 473
Equity / Surplus S_0	236 071 105	228 855 265
Dollar Duration Gap	-2 468 224 859	-2 603 760 247
Duration Gap	-10.4554	-11.3773
Dollar Convexity Gap	-56 766 821 365	-40 183 726 757
Convexity Gap	-240.4649	-175.5858

TABLE 1.7 – Duration and convexity gaps of the company.

The duration gap is negative in both cases. This means that liabilities are more sensitive to interest rate movements than assets. Therefore, a decrease in interest rates reduces equity. The negative convexity gap also indicates that liabilities have higher convexity than assets, making the company exposed to larger interest-rate shocks.

Question 8 : Equity Immunization with a Forward Start Swap

The objective is to hedge the negative dollar duration gap :

$$\$D_{Gap} = -2\,468\,224\,859.$$

We use a receiver forward start swap on Euribor 6M starting in 5 years and maturing in 10 years. The fixed leg pays semiannually, with $\tau = 0.5$ and payment dates :

$$T_i \in \{5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10\}.$$

The fair forward swap rate is :

$$K^* = \frac{P(0, T_0) - P(0, T_n)}{\tau \sum_{i=1}^{10} P(0, T_i)} = 4.2201\%.$$

The dollar duration per unit nominal of the receiver swap is :

$$\$D_{swap}^{unit} = K^* \tau \sum_{i=1}^{10} T_i P(0, T_i) + T_n P(0, T_n) - T_0 P(0, T_0) = 3.856098.$$

The hedge notional is chosen such that the post-hedge duration gap is zero :

$$N^* = \frac{-\$D_{Gap}}{\$D_{swap}^{unit}} = 640\,083\,611.$$

The appropriate position is therefore a **receiver swap** : the company receives the fixed leg and pays Euribor 6M. This adds positive dollar duration to offset the negative duration gap.

Quantity	Base	+25 bps	-25 bps
Asset value	1 774 391 738	1 752 626 636	1 796 479 082
Liability value	1 538 320 633	1 510 560 248	1 566 758 088
Equity without swap	236 071 105	242 066 388	229 720 994
Change in equity without swap	–	5 995 283	-6 350 111
% change without swap	–	2.54%	-2.69%
Swap value	0	-6 057 284	6 286 041
Equity with swap	236 071 105	236 009 105	236 007 035
Change in equity with swap	–	-62 000	-64 070
% change with swap	–	-0.03%	-0.03%

TABLE 1.8 – Exact equity impact of a ± 25 bps parallel shift.

Without the swap, equity is materially exposed to interest-rate movements. A +25 bps shock increases equity by 5.995 M€, while a –25 bps shock decreases it by 6.350 M€. After adding the receiver swap, the impact is reduced to approximately –0.06 M€ in both scenarios. The hedge is therefore effective at neutralizing the first-order interest-rate exposure. The small residual loss is due to convexity effects.

Question 9 : Risk Limits in Terms of Duration Gap

The company does not want to lose more than 15% of its equity under an 80 bps parallel shift. Using the first-order approximation :

$$\Delta S \approx -D_{Gap} S_0 \Delta r,$$

the risk constraint is :

$$|D_{Gap}| S_0 |\Delta r| \leq 0.15 S_0.$$

Thus :

$$|D_{Gap}| \leq \frac{0.15}{0.008} = 18.75.$$

The admissible interval is therefore :

$$D_{Gap} \in [-18.75, 18.75].$$

The corresponding dollar duration limit is :

$$|D_{Gap}^{limit}| = 18.75 \times S_0 = 4\,426\,333\,220.$$

Indicator	Current Value	Limit	Within Limits ?
Duration Gap	-10.4554	± 18.75	Yes
Dollar Duration Gap	-2 468 224 859	$\pm 4 426 333 220$	Yes

TABLE 1.9 – Risk limits in terms of duration gap.

The current duration gap is within the admissible interval :

$$|D_{Gap}| = 10.4554 < 18.75.$$

The company is therefore within the interest-rate risk limit. No immediate remediation is required. If the limit were breached, possible actions would include entering into swaps, increasing long-duration assets, or rebalancing the bond portfolio.

Question 10 : Credit Spread Durations

Theoretical Framework

The value of a coupon bond discounts promised cash flows at the sum of the risk-free rate and a credit spread :

$$V_0^a = \sum_{i=1}^n a_i e^{-\left(r_0^c(t_i) + s_0^c(t_i)\right)t_i}.$$

The dollar credit spread duration is defined as :

$$\$D_0^{CS} = -\frac{\partial V_0^a}{\partial s_0} = \sum_{i=1}^n t_i a_i e^{-\left(r_0^c(t_i) + s_0^c(t_i)\right)t_i}.$$

Since r_0 and s_0 enter the discount exponent symmetrically, a parallel shift of either produces the same first-order price impact. As established in the course (Part II, Chapter 4), the interest rate and credit spread durations are therefore identical :

$$\boxed{D_0^{CS} = D_0^{IR} \quad \text{and} \quad \$D_0^{CS} = \$D_0^{IR}.}$$

Credit Spread Duration of the Technical Provisions

The technical provisions are discounted at risk-free Nelson–Siegel rates only ; no credit spread enters the liability valuation. Consequently :

$$D_{TP}^{CS} = 0 \quad \text{and} \quad \$D_{TP}^{CS} = 0.$$

Credit Spread Duration of Equity

Since $S_0 = V_0^a - V_0^l$ and $\$D_l^{CS} = 0$:

$$\$D_S^{CS} = \$D_a^{CS}, \quad D_S^{CS} = \frac{\$D_a^{CS}}{S_0} = D_a^{CS} \cdot \frac{V_0^a}{S_0}.$$

Numerical Results

TABLE 1.10 – Credit spread durations – bonds, asset portfolio, technical provisions and equity.

Instrument	Market Value (€)	$\$D^{\text{IR}}$ (€)	D^{IR} (yr)	D^{CS} (yr)
Bond 1	518 430 067	1 517 652 932	2.9274	2.9274
Bond 2	588 515 755	2 743 706 377	4.6621	4.6621
Bond 3	667 445 916	4 508 775 676	6.7553	6.7553
Asset Portfolio	1 774 391 738	8 770 134 986	4.9426	4.9426
Technical Provisions	1 538 320 633	11 238 359 845	7.3056	0
Equity	236 071 105	−2 468 224 859	−10.455	37.150

Interpretation

For the three bonds, $D^{CS} = D^{\text{IR}}$. The technical provisions carry zero credit spread sensitivity since they are valued at risk-free rates. A credit spread widening that reduces asset values leaves liabilities unchanged, so the full impact falls on equity.

Question 11 : Dollar Key Rate Durations

Theoretical Framework

Key rate durations decompose the total interest rate risk into exposures at five specific maturities : $\text{KR}_0 = 0\text{Y}$, $\text{KR}_1 = 5\text{Y}$, $\text{KR}_2 = 10\text{Y}$, $\text{KR}_3 = 15\text{Y}$, $\text{KR}_4 = 20\text{Y}$. The zero-coupon rate at any maturity t is obtained by piecewise-linear interpolation between adjacent key rates :

$$r_0^c(t) = w(t, t_k, t_{k+1}) \text{KR}_k + (1 - w(t, t_k, t_{k+1})) \text{KR}_{k+1}, \quad w = \frac{t_{k+1} - t}{t_{k+1} - t_k}, \quad t \in [t_k, t_{k+1}).$$

The dollar key rate duration of a cash flow a_i at time t_i with respect to KR_k is :

$$\$D_i^{\text{KR}_k} = t_i \cdot \text{PV}(a_i) \cdot \frac{\partial r_0^c(t_i)}{\partial \text{KR}_k}.$$

The total dollar KRD of a portfolio is the sum across all cash flows. As a verification, the sum across all key rates must equal the total \$Duration :

$$\sum_k \$D^{KR_k} = \$D.$$

Numerical Results : Asset Portfolio

All bond cash flows fall within the [0Y, 10Y] range, so $\$D^{KR_{15}} = \$D^{KR_{20}} = 0$ for all assets.

TABLE 1.11 – Dollar key rate durations – bonds and asset portfolio (€).

Instrument	$\$D^{KR0}$	$\$D^{KR5}$	$\$D^{KR10}$	$\$D^{KR15}$	$\$D^{KR20}$	Sum = \$D
Bond 1 (3Y)	617 022 850	900 630 082	0	0	0	1 517 652 932
Bond 2 (5Y)	77 161 864	2 666 544 513	0	0	0	2 743 706 377
Bond 3 (8Y)	122 090 126	2 029 021 298	2 357 664 253	0	0	4 508 775 676
Asset Portfolio	816 274 840	5 596 195 893	2 357 664 253	0	0	8 770 134 986

Numerical Results : Technical Provisions

Each model point has a single cash flow at its contract maturity T_i , so $\$D_i^{KR_k} = EV_i \cdot T_i \cdot \partial r_0^c(T_i) / \partial KR_k$. Maturities range from 1 to 13 years, generating exposure across KR0 through KR15.

TABLE 1.12 – Dollar key rate durations – technical provisions (€).

Group (T)	$\$D^{KR0}$	$\$D^{KR5}$	$\$D^{KR10}$	$\$D^{KR15}$	$\$D^{KR20}$	Sum
G7 ($T = 1Y$)	193 244 370	48 311 092	0	0	0	241 555 462
G6 ($T = 3Y$)	212 077 150	318 115 724	0	0	0	530 192 874
G5 ($T = 5Y$)	0	874 513 694	0	0	0	874 513 694
G4 ($T = 7Y$)	0	868 574 100	579 049 400	0	0	1 447 623 500
G3 ($T = 9Y$)	0	472 480 529	1 889 922 115	0	0	2 362 402 644
G2 ($T = 11Y$)	0	0	1 776 871 687	444 217 922	0	2 221 089 609
G1 ($T = 13Y$)	0	0	1 424 392 825	2 136 589 237	0	3 560 982 062
Tech. Prov.	405 321 519	2 581 995 139	5 670 236 027	2 580 807 159	0	11 238 359 845

Interpretation

The asset portfolio is concentrated at KR5 ($\$D^{KR5} / \$D = 5,596,195,893 / 8,770,134,986 = 63.8\%$ of total \$Duration), reflecting the intermediate maturities of the three bonds. It has no exposure beyond KR10.

The technical provisions, by contrast, carry significant exposure at KR10 and KR15, driven by the long-dated model points (groups G1–G3 with maturities 9–13 years). This structural mismatch, assets concentrated at KR5, liabilities at KR10 and KR15, is the source of the duration gap identified in Question 7 and directly motivates the hedging analysis in Questions 13 and 14.

Question 12 : Dollar Key Rate Durations of the Equity

Methodology

The dollar key rate duration of equity is obtained by subtracting the liability \$KRD from the asset \$KRD at each key rate :

$$\$D_S^{KR_k} = \$D_a^{KR_k} - \$D_l^{KR_k}, \quad D_S^{KR_k} = \frac{\$D_S^{KR_k}}{S_0}.$$

As a verification, the sum across all key rates equals the total \$Duration gap from Question 7 : $\sum_k \$D_S^{KR_k} = -2,468,224,859 \text{ €}$.

Numerical Results

TABLE 1.13 – Dollar key rate durations of the equity (€).

Key Rate	$\$D_k^a$ (€)	$\$D_k^l$ (€)	$\$D_k^s$ (€)	D_k^s (yr)
KR0 (0Y)	+816 274 840	+405 321 519	+410 953 321	+1.74
KR5 (5Y)	+5 596 195 893	+2 581 995 139	+3 014 200 753	+12.77
KR10 (10Y)	+2 357 664 253	+5 670 236 027	−3 312 571 774	−14.03
KR15 (15Y)	0	+2 580 807 159	−2 580 807 159	−10.93
KR20 (20Y)	0	0	0	0
Sum	+8 770 134 986	+11 238 359 845	−2 468 224 859	−10.46

Interpretation

The equity \$KRD is positive at KR5 and negative at KR10 and KR15. This means that a rise in the 5Y rate increases equity value, while a rise in the 10Y or 15Y rate reduces it. The negative gaps at KR10 and KR15 arise because liabilities have longer maturities than assets : the asset portfolio has no exposure beyond KR10, whereas groups G1 and G2 of the technical provisions mature at 13 and 11 years. The company is therefore not exposed to a simple parallel shift of the yield curve, but to changes in its **shape**, in particular any movement that widens the spread between long and short rates.

Question 13 : First-Order Impact of Simultaneous Key Rate Changes

Methodology

Using the dollar key rate durations of equity from Question 12, the first-order approximate change in equity value from simultaneous key rate shifts is :

$$\Delta S_0 \approx - \sum_k \$D_S^{KR_k} \cdot \Delta_k.$$

Results

TABLE 1.14 – First-order impact on equity of simultaneous key rate shifts.

Key Rate	Shift Δ_k	$\$D_k^S$ (€)	$-\$D_k^S \cdot \Delta_k$ (€)
KR0 (0Y)	0 bps	+410 953 321	0
KR5 (5Y)	+15 bps	+3 014 200 753	−4 521 301
KR10 (10Y)	−15 bps	−3 312 571 774	−4 968 858
KR15 (15Y)	−25 bps	−2 580 807 159	−6 452 018
KR20 (20Y)	0 bps	0	0
Total			−15 942 177

Interpretation

The approximate impact on equity is $\Delta S_0 \approx -\text{€}15.94 \text{ M}$, or $-15,942,177 / 236,071,105 = -6.75\%$ of current equity. All three active shifts are adverse : the rise in KR5 reduces equity because assets are more sensitive than liabilities at that maturity, while the falls in KR10 and KR15 increase liability values with no offsetting effect on the asset side, which has zero exposure at those maturities.

Question 14 : Forward Start Swap Hedging by Key Rate

Instrument

The hedging instrument is the forward start swap from Question 8 : Euribor 6M reference, start date $T_0 = 5Y$, maturity $T_n = 10Y$, semi-annual floating payments, fixed rate $K^* = 4.2201\%$. Since all cash flows fall between years 5 and 10, the swap has non-zero \$KRD only at KR5 and KR10. The dollar key rate durations per unit nominal are :

TABLE 1.15 – Dollar key rate durations of the swap per unit nominal.

Key Rate	$\$D_k^{\text{fixed}}$	$\$D_k^{\text{float}}$	$\$D_k^{\text{swap}}$
KR0 (0Y)	0	0	0
KR5 (5Y)	0.5047	4.2282	-3.7234
KR10 (10Y)	7.5795	0	+7.5795
KR15 (15Y)	0	0	0
KR20 (20Y)	0	0	0
Sum	8.0843	4.2282	3.8561

The floating leg matures at $T_0 = 5Y$, which falls exactly on KR5, so $\$D_{KR5}^{\text{float}} = 5 \times P(0, 5) = 4.2282$ and all other floating leg KRDs are zero.

Sub-question A : Immunisation Against KR5

The hedge condition requires the total KR5 exposure of equity plus N units of the swap to equal zero :

$$\$D_{KR5}^S + N \cdot \$D_{KR5}^{\text{swap}} = 0 \implies N = -\frac{\$D_{KR5}^S}{\$D_{KR5}^{\text{swap}}} = -\frac{3,014,200,753}{-3.7234} = +809,519,484 \text{ €}.$$

Since $N > 0$, the company must enter a **payer swap** (pay fixed, receive floating). This is intuitive : the equity has a positive KR5 gap, meaning it loses value when the 5Y rate rises. A payer swap gains value when rates rise, offsetting this exposure.

Sub-question B : Immunisation Against KR15

The swap matures at $T_n = 10Y$ and has no cash flows beyond year 10, so its KR15 sensitivity is identically zero :

$$\$D_{KR15}^{\text{swap}} = 0.$$

The hedge condition $N \times 0 = -\$D_{KR15}^S$ has no finite solution. Immunisation against KR15 is therefore **impossible** with this instrument. To hedge the KR15 exposure of -2,580,807,159 €, a different instrument would be needed, such as a swap starting at year 10 and maturing at year 15, or a 15-year zero-coupon bond.

Chapitre 2

Stochastic Model

Question 1-a : Martingale Properties under Q

The theoretical values of the two mathematical expressions are equal to :

$$\boxed{E^Q \left[\frac{1}{B_t} \frac{S_t}{S_0} \middle| \mathcal{F}_0 \right] = 1 \quad \text{and} \quad E^Q \left[\frac{1}{B_t} \frac{P(t, T)}{P(0, T)} \middle| \mathcal{F}_0 \right] = 1}$$

This property follows from the *absence of arbitrage opportunities* assumption in financial markets. According to the fundamental theorem of asset pricing, there exists an equivalent probability measure Q such that the discounted price of any tradable asset is a Q -martingale. Let X_t be the price of a financial asset at time t . The discounted process $\tilde{X}_t$ defined by :

$$\tilde{X}_t = \frac{X_t}{B_t}$$

is a martingale under Q . By definition of the martingale property, the conditional expectation at time 0 of the future discounted price is equal to the current discounted price :

$$E^Q \left[\frac{X_t}{B_t} \middle| \mathcal{F}_0 \right] = \frac{X_0}{B_0}$$

Since the money market account is initialized at $B_0 = 1$, we have :

$$E^Q \left[\frac{X_t}{B_t} \middle| \mathcal{F}_0 \right] = X_0$$

By dividing both sides of the equation by X_0 (to normalize the initial price to 1), we obtain :

$$E^Q \left[\frac{1}{B_t} \frac{X_t}{X_0} \middle| \mathcal{F}_0 \right] = 1$$

Application to the two cases

The mutual fund S_t and the zero-coupon bond $P(t, T)$ are tradable asset. Therefore, $\frac{S_t}{B_t}$ and $\frac{P(t, T)}{B_t}$ are martingale. Then :

$$E^Q \left[\frac{1}{B_t} \frac{S_t}{S_0} \middle| \mathcal{F}_0 \right] = 1 \quad \text{and} \quad E^Q \left[\frac{1}{B_t} \frac{P(t, T)}{P(0, T)} \middle| \mathcal{F}_0 \right] = 1$$

Question 1-b : Monte-Carlo Estimation of Martingale Properties

To estimate the theoretical expectations

$$\mathbb{E}^Q \left[\frac{1}{B_t} \frac{S_t}{S_0} \middle| \mathcal{F}_0 \right] \quad \text{and} \quad \mathbb{E}^Q \left[\frac{1}{B_t} \frac{P(t, T)}{P(0, T)} \middle| \mathcal{F}_0 \right],$$

we implement a Monte-Carlo simulation under the risk-neutral measure Q .

Simulation Scheme

For a time step Δt , we discretize the dynamics as follows :

- (1) **Correlated Brownian motions** : Generate two independent standard normals $Z_1, Z_2 \sim \mathcal{N}(0, 1)$ and construct

$$dW_t^{Q,1} = \sqrt{\Delta t} Z_1, \quad dW_t^{Q,2} = \sqrt{\Delta t} (\rho Z_1 + \sqrt{1 - \rho^2} Z_2).$$

- (2) **Hull-White factor** : Update x_t via

$$x_{t+\Delta t} = x_t - a_r x_t \Delta t + \sigma_r dW_t^{Q,1}, \quad x_0 = 0.$$

- (3) **Short rate** : Compute $r_t = x_t + \phi(t)$, where $\phi(t)$ is calibrated to the initial Nelson-Siegel curve.

- (4) **Money market account** :

$$B_{t+\Delta t} = B_t \exp(r_t \Delta t), \quad B_0 = 1.$$

- (5) **Mutual fund** :

$$S_{t+\Delta t} = S_t \exp \left((r_t - \frac{1}{2} \sigma_S^2) \Delta t + \sigma_S dW_t^{Q,2} \right), \quad S_0 = 1.$$

- (6) **Zero-coupon bond** : For maturity $T \geq t$,

$$P(t, T) = \exp \left(-r_0^{NS} (T - t) \cdot (T - t) \right),$$

Estimators

For each trajectory $j = 1, \dots, N$, compute at horizon t :

$$M_S^{(j)}(t) = \frac{1}{B_t^{(j)}} \frac{S_t^{(j)}}{S_0}, \quad M_P^{(j)}(t, T) = \frac{1}{B_t^{(j)}} \frac{P(t, T)^{(j)}}{P(0, T)}.$$

The Monte-Carlo estimators computed for $N = 1000$ are then :

$$\widehat{\mathbb{E}}[M_S(t)] = \frac{1}{N} \sum_{j=1}^N M_S^{(j)}(t), \quad \widehat{\mathbb{E}}[M_P(t, T)] = \frac{1}{N} \sum_{j=1}^N M_P^{(j)}(t, T).$$

Horizon t	$\widehat{\mathbb{E}}[\text{Ms}]$	$\widehat{\mathbb{E}}[\text{Mp}]$	std_ms	std_mp
t = 1 an	1.008	0.985	0.204	0.130
t = 5 ans	1.009	0.988	0.524	0.280
t = 10 ans	1.013	1.046	0.801	0.362

TABLE 2.1 – Monte-Carlo simulation

Question 2 : Economic Value of Technical Provisions

General Formula

$$V_0^{TP} = \mathbb{E}^{\mathbb{Q}} \left[\sum_{i=1}^7 \frac{N_i \cdot Cl_i^{\text{final}} \cdot \mathbb{1}_{\{T_i\text{-survival}\}}}{B_{T_i}} \right]$$

where :

- N_i : Notional of contract i
- Cl_i^{final} : Final guaranteed capital
- $\mathbb{1}_{\{T_i\text{-survival}\}}$: Survival indicator using Makeham model and Bernouilli simulation
- $B_{T_i} = \exp\left(\int_0^{T_i} r_s ds\right)$: Money market account

Case 1 : Without Profit-Sharing

$$Cl_i^{\text{final}} = Cl_i \quad \text{constant guaranteed capital}$$

Case 2 : With Profit-Sharing

At each year $k = 1, \dots, T_{\max}$:

$$\begin{aligned} r_A(k) &= \frac{V_k^a - V_{k-1}^a}{V_{k-1}^a} \quad (\text{asset portfolio return}) \\ r_i^{PS}(k) &= \min(0.85 \cdot \max(r_A(k) - 0.001 - g_i, 0), 0.03) \\ Cl_i(k) &= Cl_i(k-1) \times (1 + r_i^{PS}(k)) \end{aligned}$$

with $Cl_i^{\text{final}} = Cl_i(T_i)$.

Asset Portfolio Dynamics

$$\frac{dV_t^a}{V_t^a} = [0.05r_t + 0.15(r_t) + 0.80\mu_P(t)] dt + 0.15\sigma_S dW_t^{Q,2} + 0.80\sigma_P(t) dW_t^{Q,P} \quad (2.1)$$

where $\mu_P(t)$ and $\sigma_P(t)$ are the drift and volatility of the 4-year zero-coupon bond under $\mathbb{Q}$.

The table 2.2 below reports the economic value of technical provisions under two scenarios. Introducing the profit-sharing mechanism increases V_0^{TP} by **6.4%**, reflecting the value of the embedded bonus option granted to policyholders. Consequently, the initial equity $S_0 = A_0 - V_0^{TP}$ declines from 0.634 to 0.535 Md€, a reduction of **15.6%**. The implicit price of the profit-sharing option, defined as $\Delta V_0^{TP} = V_0^{TP,PS} - V_0^{TP,noPS}$, amounts to **0.098 Md€**. This quantifies the trade-off between policyholder benefits and shareholder capital in the presence of performance-linked guarantees.

Scenario	Technical Provisions V_0^{TP}	Equity	PS Cost
Without Profit-Sharing (Md€)	1.536	0.634	–
With Profit-Sharing (Md€)	1.635	0.535	0.098

TABLE 2.2 – Economic value of technical provisions and equity

Question 3 :Duration of Technical Provisions

Scenario	Duration D^{num} (years)
Without Profit-Sharing	7.30
With Profit-Sharing	7.23

TABLE 2.3 – Numerical duration of technical provisions ($\varepsilon = 5$ bps).

The table 2.3 shows that the profit-sharing mechanism **slightly reduces** the duration of technical provisions from 7.30 to 7.23 years, a **1.1%** decrease. This occurs because the bonus rate $r^{PS}(k)$ acts as a cushion : when interest rates rise, asset returns tend to increase, triggering higher bonuses that partially offset the discounting effect on liabilities. Conversely, when rates fall, bonuses decrease, limiting the increase in liability values. The profit-sharing option thus **moderately dampens** the interest rate sensitivity of the guaranteed benefits, resulting in a marginally lower effective duration. These results are consistent with the deterministic duration computed in Part I (≈ 7.3 years), confirming the robustness of the numerical implementation.

Question 4 : Duration Gap and Impact on Equity

Asset Duration Calculation

The asset portfolio consists of 5% cash, 15% mutual fund, and 80% 4-year zero-coupon bonds. The durations are :

- Cash : $D_{\text{cash}} = 0$
- Mutual fund : $D_{\text{fund}} = 0$
- Rolling ZC bond : $D_{\text{ZC}} = B(0, 4) = 3.20$ years

The weighted portfolio duration is :

$$D^{\text{Assets}} = 0.05 \times 0 + 0.15 \times 0 + 0.80 \times 3.20 = \mathbf{3.20} \text{ years.}$$

The gap duration is the leverage-adjusted weighted duration :

$$D^{\text{Equity}} = \frac{A_0}{S_0} D^A - \frac{L_0}{S_0} D^L$$

This measures the sensitivity of shareholder capital to interest rate changes, accounting for the fact that equity is a residual claim.

Duration Gap	Dur.years
Without PS : D^{Equity}	-6.75
With PS : D^{Equity}	-9.11

TABLE 2.4 – Duration of equity analysis with and without profit-sharing.

The negative duration of equity indicates that shareholder capital **decreases** when interest rates fall. The profit-sharing mechanism **amplifies** this sensitivity from -6.75 to -9.11 years because it increases the market value of liabilities while only marginally reducing their duration, thereby increasing the leverage effect on equity.

Impact of a 20bps Downward Parallel Shift

Component	Without PS (M€)	With PS (M€)
ΔA	+13.888	
ΔL	+22.4256	+23.6421
ΔS_0	-8.537	-9.754
Via Duration of Equity		
$\Delta S_0 \approx -D^{\text{Eq}} \times S_0 \times \Delta r$	-8.557	-9.745

TABLE 2.5 – Impact on equity of a downward parallel shift in interest rates.

For a downward shift of $\Delta r = -0.002$:

- **Equity impact** : Both methods yield consistent results :
 - Without PS : $\Delta S_0 \approx -8.55$ M€
 - With PS : $\Delta S_0 \approx -9.75$ M€

The close agreement between $\Delta A - \Delta L$ and $-D^{\text{Equity}} \times S_0 \times \Delta r$ validates the numerical consistency of the duration estimates.

Interpretation :

- (1) The duration of equity is **negative** : falling rates erode shareholder capital.
- (2) Profit-sharing **increases** sensitivity, making equity more vulnerable to rate drops.
- (3) This occurs because PS increases liability values (higher L_0/S_0) while only slightly reducing duration, amplifying the leverage effect.

Question 5 : Explicit Delta

The explicit delta measures the first-order sensitivity of the balance sheet to the Hull–White risk factor x_0 . For a zero-coupon bond, we use :

$$\frac{\partial P(0, T)}{\partial x_0} = -B(0, T)P(0, T), \quad B(0, T) = \frac{1 - e^{-a_r T}}{a_r}.$$

The asset delta comes only from the zero-coupon bond allocation :

$$\Delta_A = -x_{\text{bond}} V_0^A B(0, 4) = -5\,723\,244\,000.82.$$

For the liabilities without profit-sharing, the delta is :

$$\Delta_{TP}^{noPS} = \sum_i N_i C l_i(T_i)_{T_i} p_{x_i} [-B(0, T_i)P(0, T_i)].$$

The total liability delta is :

$$\Delta_{TP}^{noPS} = -7\,314\,623\,984.71.$$

The equity delta is therefore :

$$\Delta_S^{noPS} = \Delta_A - \Delta_{TP}^{noPS} = 1\,591\,379\,983.89.$$

Component	Delta w.r.t. x_0 (€)
Assets	-5 723 244 000.82
Technical provisions without PS	-7 314 623 984.71
Equity without PS	1 591 379 983.89

TABLE 2.6 – Explicit delta of assets, liabilities and equity.

The positive equity delta means that an increase in x_0 increases equity. This is because the liabilities are more sensitive to interest rates than the assets; when rates rise, liabilities decrease more than assets.

Question 6 : Numerical Delta with Profit-Sharing

With profit-sharing, the liability cash flows depend on the simulated asset returns. Therefore, the delta is computed numerically using a centered finite difference :

$$\Delta_{TP}^{PS} = \frac{TP(x_0 + h) - TP(x_0 - h)}{2h}, \quad h = 10^{-4}.$$

The numerical result is :

$$\Delta_{TP}^{PS} = -7\,385\,503\,934.77.$$

Using the same asset delta as in Question 5 :

$$\Delta_S^{PS} = \Delta_A - \Delta_{TP}^{PS} = 1\,662\,259\,933.95.$$

Quantity	Value (€)
$TP(x_0 + h)$	1 634 200 765.31
$TP(x_0 - h)$	1 635 677 866.10
ΔTP^{PS}	-7 385 503 934.77
ΔV_A	-5 723 244 000.82
ΔS^{PS}	1 662 259 933.95

TABLE 2.7 – Numerical delta with profit-sharing.

Profit-sharing slightly increases the interest-rate sensitivity of the liabilities. The difference with the no profit-sharing case is :

$$\Delta_{TP}^{PS} - \Delta_{TP}^{noPS} = -70\,879\,950.06.$$

Question 7 : Forward Start Swap Hedging Strategy

To hedge the equity delta, we use a forward start receiver swap on Euribor 6M, starting at $T_0 = 5$ years and maturing at $T_n = 10$ years. The payment frequency is semiannual, hence $\tau = 0.5$.

The fair forward swap rate is :

$$K^* = \frac{P(0, T_0) - P(0, T_n)}{\tau \sum_i P(0, T_i)} = 4.2201\%.$$

The swap delta per unit nominal is :

$$\Delta_{swap} = -(\Delta_{CB} - \Delta_{FR}) = -1.856314.$$

The hedge notional is chosen such that :

$$\Delta_S + N^* \Delta_{swap} = 0.$$

Therefore :

$$N^* = -\frac{\Delta_S}{\Delta_{swap}}.$$

Quantity	Value
Forward swap rate K^*	4.2201%
Δ_{CB}	5.183615
Δ_{FR}	3.327301
Δ_{swap}	-1.856314
N^* without PS	857 279 587.53 €
N^* with PS	895 462 758.71 €
Swap position	Receiver swap

TABLE 2.8 – Forward start swap hedge.

The optimal position is a **receiver swap**, i.e. the company receives the fixed leg and pays Euribor 6M. This position offsets the positive equity delta and immunizes the company against first-order interest-rate movements.

Question 8 : Delta with respect to the Mutual Fund Initial Value

The sensitivity to the initial mutual fund value S_0 is computed with a centered finite difference using a 1% shock :

$$\Delta_{TP}^{S_0} = \frac{TP(S_0 + h_S) - TP(S_0 - h_S)}{2h_S}, \quad h_S = 0.01S_0.$$

The numerical results show :

$$TP(S_0 + h_S) = TP(S_0 - h_S) = 1\,634\,939\,070.89.$$

Thus :

$$\Delta_{TP}^{S_0} = 0.$$

The asset portfolio sensitivity is :

$$\Delta_A^{S_0} = 325\,500\,000,$$

and therefore :

$$\Delta_S^{S_0} = 325\,500\,000.$$

Quantity	Value (€)
$\Delta_{TP}^{S_0}$	0
$\Delta_A^{S_0}$	325 500 000
$\Delta_S^{S_0}$	325 500 000

TABLE 2.9 – Delta with respect to the mutual fund initial value.

The technical provisions are locally insensitive to S_0 . The equity sensitivity comes entirely from the asset side.

Question 9 : Vega of the Technical Provisions

The vega measures the sensitivity of technical provisions to the mutual fund volatility σ_S :

$$Vega_{TP} = \frac{TP(\sigma_S + h_\sigma) - TP(\sigma_S - h_\sigma)}{2h_\sigma}.$$

Using $h_\sigma = 0.001$, the numerical result is :

$$Vega_{TP} = 62\,935\,507.95.$$

Since the asset portfolio value at time 0 does not directly depend on σ_S :

$$Vega_A = 0, \quad Vega_S = -Vega_{TP}.$$

Quantity	Value (€)
$TP(\sigma_S + h_\sigma)$	1 635 002 086.16
$TP(\sigma_S - h_\sigma)$	1 634 876 215.14
$Vega_{TP}$	62 935 507.95
$Vega_S$	-62 935 507.95

TABLE 2.10 – Vega of the technical provisions.

The vega of the technical provisions is positive because profit-sharing behaves like an embedded option on asset returns. Higher volatility increases the probability of high future bonuses, which increases the value of liabilities and decreases equity.

Question 10 : Distribution under the Real-World Measure

Under the real-world measure $\mathbb{P}$, the Hull–White factor satisfies :

$$dx_t = -a_r x_t dt + \sigma_r \lambda_1 dt + \sigma_r dW_t^{\mathbb{P}}.$$

Thus, x_1 is normally distributed with :

$$\mathbb{E}[x_1] = x_0 e^{-a_r} + \frac{\sigma_r \lambda_1}{a_r} (1 - e^{-a_r}), \quad \text{Var}(x_1) = \frac{\sigma_r^2}{2a_r} (1 - e^{-2a_r}).$$

The mutual fund follows :

$$\frac{dS_t}{S_t} = (r_t + \sigma_S \lambda_2) dt + \sigma_S dW_t^{\mathbb{P}}.$$

Therefore, S_1 is lognormally distributed.

Quantity	Value
$\mathbb{E}[x_1]$	-0.003414
$Std(x_1)$	0.016660
$q_{10\%}(x_1)$	-0.024765
$q_{90\%}(x_1)$	0.017937
$q_{10\%}(r_1)$	0.0016
$q_{90\%}(r_1)$	0.0443
$\mathbb{E}[S_1]$	1.07842
Expected return	7.84%
$Std(S_1)$	0.21786
$q_{10\%}(S_1)$	0.81807
$q_{90\%}(S_1)$	1.36590

TABLE 2.11 – One-year distributions under $\mathbb{P}$.

The short-rate factor has a negative expected value over one year due to the market price of interest-rate risk. The mutual fund has an expected one-year return of 7.84%, but also significant volatility, with a 10% quantile of -18.19% and a 90% quantile of $+36.59\%$.

Question 11 : risk limits in terms of delta

The management imposes the following risk tolerance constraint : the company should not lose more than 15% of its equity with a probability of 10%, due to interest rate risk. Formally :

$$\mathbb{P}^P(\Delta S_0 \leq -0.15 \times S_0) \leq 10\%$$

Using a first-order Taylor approximation, the change in equity induced by a move Δx_1 of the Hull-White factor is :

$$\Delta S_0 \approx \Delta_{x_0}^{S_0} \times \Delta x_1$$

where $\Delta_{x_0}^{S_0}$ is the equity delta computed in Question 6. Since $\Delta_{x_0}^{S_0} > 0$, equity loses value when x_1 falls. The worst-case scenario at the 10% level corresponds to the 10th percentile of x_1 under the real-world measure $\mathbb{P}$, derived in Question 10 :

$$\Delta x_1^{\text{worst}} = q_{10\%}^{x_1} = \mu_{x_1} + \sigma_{x_1} \cdot \Phi^{-1}(0.10)$$

Imposing that equity does not lose more than 15% in this scenario gives :

$$\Delta_{x_0}^{S_0} \times q_{10\%}^{x_1} \geq -0.15 \times S_0$$

Since $q_{10\%}^{x_1} < 0$, this yields the following risk limit on the absolute value of the equity delta :

$$\boxed{|\Delta_{x_0}^{S_0}| \leq \frac{0.15 \times S_0}{|q_{10\%}^{x_1}|}}$$

The numerical results are reported in the following table :

Equity with profit-sharing S_0	535,060,929 €
10th percentile of x_1 under $\mathbb{P}$	-0.024765
Current equity delta $\Delta_{x_0}^{S_0}$	1,662,259,934 €
Maximum allowable delta	3,240,815,737 €

TABLE 2.12 – Risk limit on the equity delta (interest rate risk)

The current equity delta of 1.66 Md€ is well below the maximum allowable delta of 3.24 Md€. The company is therefore within its stated risk tolerance : even in the worst 10% interest rate scenario, the loss on equity would not exceed 15% of its current value.

Question 12 : Approximate Distribution of the Profit and Loss

We compute the approximate distribution of the P&L of the company over a one-year horizon. Assuming that time does not pass and using a first-order Taylor expansion, the P&L is approximated as :

$$P\&L \approx \Delta_{x_0}^{S_0} \cdot \Delta x_1 + \Delta_{S_0}^{S_0} \cdot \Delta S_1$$

where $\Delta x_1 = x_1 - x_0$ and $\Delta S_1 = S_1 - S_0$ are the one-year moves of the two risk factors under the real-world measure $\mathbb{P}$, and $\Delta_{x_0}^{S_0}$, $\Delta_{S_0}^{S_0}$ are the equity deltas computed in Questions 6 and 8 respectively.

We simulate 100,000 scenarios of (x_1, S_1) under $\mathbb{P}$ using the analytical distributions derived in Question 10 :

$$x_1 \sim \mathcal{N}(\mu_{x_1}, \sigma_{x_1}^2)$$

$$\ln S_1 \sim \mathcal{N}\left(\left(\varphi(0) + \sigma_S \lambda_2 - \frac{1}{2}\sigma_S^2\right), \sigma_S^2\right)$$

For each scenario $s = 1, \dots, 100,000$, the P&L is computed via the linear approximation and the empirical distribution is constructed from these 100,000 values. The results are reported in the following table :

Mean P&L	19,957,999 €
Std P&L	76,416,914 €
Percentile 0.5%	−146,273,350 €
Percentile 99.5%	256,929,798 €

TABLE 2.13 – Approximate P&L distribution over a one-year horizon

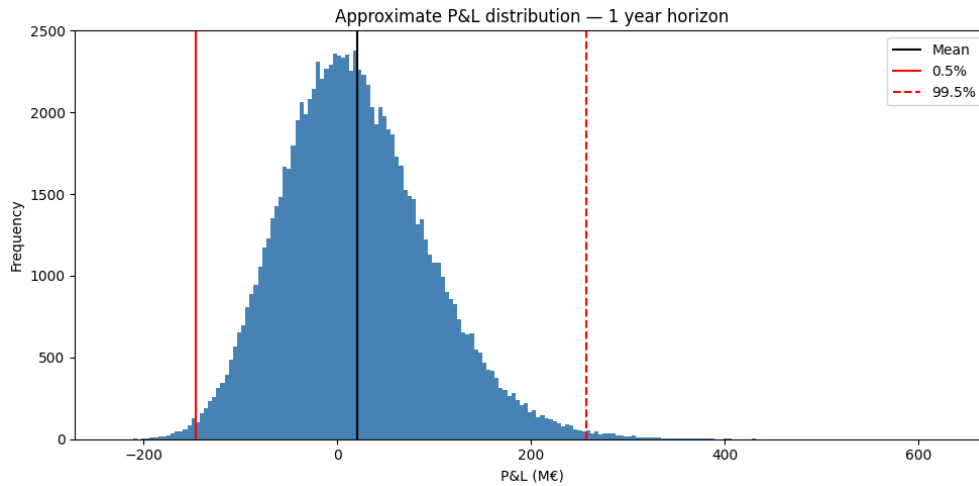


FIGURE 2.1 – Approximate P&L distribution — 1 year horizon

The distribution exhibits a positive mean of 19.96 M€, reflecting the positive risk premiums on both interest rates and equity under the real-world measure $\mathbb{P}$. The distribution is right-skewed, driven by the log-normal component of ΔS_1 : the upside from the mutual fund is unbounded while the downside is floored. The asymmetry between the 0.5% percentile (−146.3 M€) and the 99.5% percentile (+256.9 M€) confirms this. The standard deviation of 76.4 M€ represents approximately 14.3% of the current equity, indicating a material but manageable level of risk.

Question 13 : Approximate Distribution of the Equity

Based on the P&L distribution computed in Question 12, the approximate distribution of the equity of the company over a one-year horizon is obtained by adding the

current equity S_0 to each simulated P&L scenario :

$$S_1^{(s)} = S_0 + P\&L^{(s)}$$

where $S_0 = V_0^A - V_0^{TP,PS} = 535,060,929$ € is the current equity with profit-sharing. Since this is a simple shift of the P&L distribution, the standard deviation is unchanged while the mean shifts upward by S_0 .

Mean Equity	555,018,928 €
Std Equity	76,416,914 €
Percentile 0.5%	388,787,580 €
Percentile 99.5%	791,990,727 €

TABLE 2.14 – Approximate equity distribution over a one-year horizon

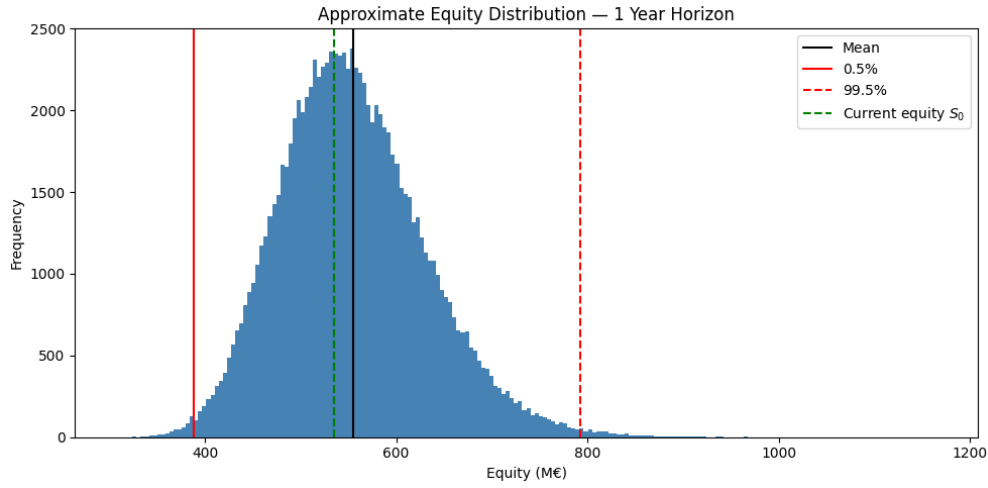


FIGURE 2.2 – Approximate equity distribution — 1 year horizon

The mean equity of 555.0 M€ exceeds the current equity S_0 of 535.1 M€ by approximately 20.0 M€, consistent with the positive mean P&L from Question 12. The 0.5% percentile of 388.8 M€ represents the worst-case equity level at the 99.5% confidence level, this value will be used directly in Question 14 to compute the SCR. The right skew observed in Question 12 is preserved here, with the upside scenario (791.9 M€) being significantly further from the mean than the downside (388.8 M€).

Question 14 : SCR and Solvency Ratio

Under Solvency II, the Solvency Capital Requirement (SCR) is defined as the 99.5% Value at Risk of the loss over a one-year horizon. Using the equity distribution computed in Question 13, it is given by :

$$\text{SCR} = S_0 - q_{0.5\%}^{S_1}$$

where $q_{0.5\%}^{S_1}$ is the 0.5% percentile of the equity distribution at the one-year horizon. The solvency ratio is then defined as :

$$\text{Solvency Ratio} = \frac{S_0}{\text{SCR}}$$

A solvency ratio above 100% indicates that the company holds sufficient capital to meet its regulatory requirement.

Current equity S_0	535,060,929 €
0.5% equity percentile $q_{0.5\%}^{S_1}$	388,787,580 €
SCR	146,273,350 €
Solvency ratio	365.80%

TABLE 2.15 – SCR and solvency ratio

The SCR amounts to 146.3 M€, representing approximately $\frac{\text{SCR}}{S_0} = \frac{146,273,350}{535,060,929} = 27.3\%$ of current equity. The solvency ratio of 365.80% is well above the 100% regulatory threshold, indicating that the company is strongly capitalized relative to its risk exposure. This comfortable buffer is consistent with the findings of Question 12, where the P&L distribution showed limited downside risk relative to the equity base, and with Question 11, where the company was found to be within its delta risk limits.

Question 15 : Improving the Solvency Ratio

Diagnosis

The solvency ratio of 365.80% computed in Question 14 is already well above the 100% regulatory threshold. However, the SCR of 146.3 M€ represents approximately 27.3% of current equity, which reflects a material exposure to interest rate risk. The root cause is the structural duration mismatch identified in Question 4 : the asset portfolio holds an 80% weight in a 4-year rolling zero-coupon bond, yielding an asset duration of $D^A = 0.80 \times B(0, 4) = 3.20$ years, far below the liability duration of $D^L \approx 7.23$ years. This gap produces a large negative equity duration and a large equity delta $\Delta_{x_0}^{S_0} = +1,662,259,934$ €, which directly inflates the P&L volatility and hence the SCR.

From the first-order approximation of Question 12 :

$$P\&L \approx \Delta_{x_0}^{S_0} \cdot \Delta x_1 + \Delta_{S_0}^{S_0} \cdot \Delta S_1$$

reducing the SCR requires reducing the absolute values of the deltas, i.e. reducing the sensitivity of equity to the risk factors. **The most direct lever is to reduce the duration mismatch between assets and liabilities.**

Proposed Change

We propose replacing the 4-year rolling zero-coupon bond with a **7-year rolling zero-coupon bond**, keeping all other parameters unchanged (same weights : 5% cash, 15% mutual fund, 80% bond ; same insurance portfolio ; same profit-sharing parameters). This increases the asset duration to :

$$D_{\text{new}}^A = 0.80 \times B(0, 7) = 0.80 \times \frac{1 - e^{-0.10 \times 7}}{0.10} = 0.80 \times 5.034 = 4.027 \text{ years}$$

which is significantly closer to the liability duration of 7.23 years than the original $D^A = 3.20$ years. As a result, the duration gap narrows, the equity delta $|\Delta_{x_0}^{S_0}|$ decreases, and the P&L distribution tightens, pulling the 0.5% tail upward and reducing the SCR.

This approach is consistent with the classical ALM immunization principle discussed in the course : an insurance company should align the duration of its assets with the duration of its liabilities in order to minimize the sensitivity of its equity to interest rate movements.

Results

The following table compares the key metrics before and after the proposed change :

	Original (4Y bond)	Proposed (7Y bond)
TP with PS	1,635,060,929 €	1,650,598,399 €
Equity S_0	535,060,929 €	519,401,601 €
$\Delta_{x_0}^{S_0}$	+1,662,259,934 €	−1,227,251,510 €
$\Delta_{S_0}^{S_0}$	325,500,000 €	325,500,000 €
SCR	146,273,350 €	126,727,688 €
Solvency Ratio	365.80%	409.86%

TABLE 2.16 – Comparison of key metrics before and after the proposed asset change

Discussion

The proposed change reduces the SCR by approximately 19.5 M€ (13.3%) and improves the solvency ratio from 365.80% to 409.86%.

The equity delta $\Delta_{x_0}^{S_0}$ changes sign, from +1,662,259,934 € to −1,227,251,510 €, indicating that the 7-year bond slightly over-hedges the interest rate risk. In practice, a bond maturity between 4 and 7 years would achieve near-perfect duration matching and a near-zero equity delta, minimizing the SCR further.

The equity decreases slightly from 535.1 to 519.4 M€ because the 7-year bond has a lower initial price $P(0, 7) < P(0, 4)$, reducing asset returns in the early years and leading to slightly higher technical provisions through the PS mechanism.

It is worth noting that the equity itself decreases slightly from 535.1 to 519.4 M€ under

the new allocation. The improvement in the solvency ratio therefore stems from the SCR falling proportionally faster than the equity : the SCR decreases by $\frac{146.3-126.7}{146.3} = 13.3\%$ while the equity decreases by only $\frac{535.1-519.4}{535.1} = 3.1\%$, confirming that the dominant effect is the reduction in interest rate risk exposure rather than a change in the capital base.

Conclusion

L'asymétrie de duration entre actifs et passifs est le problème central de ce bilan. Elle produit un gap négatif qui pèse sur les fonds propres dès que les taux baissent et la couverture par receiver swap, si elle neutralise l'exposition parallèle au premier ordre, ne corrige pas les risques de déformation de la courbe que révèle l'analyse en key rate durations. La participation aux bénéfices complique le tableau : elle se comporte comme une option imbriquée dans les provisions, coûte environ 98 M€ à l'assureur et rend les fonds propres plus sensibles aux taux qu'en l'absence de mécanisme de bonus. Le SCR de 146,3 M€ et le ratio de solvabilité de 365,8% restent confortables, mais ces résultats reposent sur une approximation linéaire du P&L les risques de rachats comportementaux, de crédit et les queues épaisses de distribution conduiraient en pratique à des estimations plus sévères. La question 15 illustre bien où se situe le levier principal : allonger la duration des actifs de 3,2 à 4,0 ans suffit à réduire le SCR de 13,3%. Avant tout instrument de couverture, c'est la composition du portefeuille d'actifs qui détermine l'exposition structurelle du bilan.